Ankit Sinha

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Education

Indian Institute of Technology Patna — Bihta, Patna, India
Master of Computer Applications (MCA) — *July 2026 – 2028 (Expected)*

Lovely Professional University — Phagwara, Punjab, India
Bachelor of Computer Applications (BCA) — CGPA: 9.86 / 10.0 — *Aug 2023 – May 2026*

Technical Skills

Languages: Go, Java, Python, TypeScript, JavaScript, C++, SQL
Backend & Cloud: Gin, GORM, REST APIs, Node.js, Express.js, Next.js, AWS S3, MinIO, GCP (Cloud Run, Vertex AI)
Infrastructure: Docker, PostgreSQL, MySQL, MongoDB, Redis, RabbitMQ, Kafka, Traefik, Linux, Git
Security: JWT (HS256/RS256), OAuth 2.0, Opaque Refresh Tokens, HMAC-SHA256, CSRF Protection

Experience

Software Developer Intern — Nov 2025 – Apr 2026
Denthinkers Foundation — *Next.js 15, TypeScript, MongoDB, Razorpay, JWT*

- Shipped features across 15 API route handlers / 23 endpoints on a live production platform (denthinkers.in).
- Built the Razorpay donation module with server-side HMAC-SHA256 signature verification — INR 1.5L+ processed.
- Implemented httpOnly JWT auth with middleware-gated routes and a draft/publish CMS with slug-collision handling; tuned MongoDB connection pooling and ISR revalidation to support 2,500+ monthly visits.

Projects

AI Finance Controller | *Go, Gin, Gemini 2.5 Flash, Vertex AI, Cloud Run* — Jun 2026 – Present

- Finance reconciliation engine tying ledger, Razorpay settlement, and bank statement records into a provable chain; deployed as a 43.6 MB distroless binary on Cloud Run.
- 4-pass deterministic cascade (exact reference, date-window, batch-sum) resolves **85–95%** of records with zero API calls; Gemini is invoked only for the **~2–12%** ambiguous residue at temperature 0 with enforced `responseSchema`.
- All LLM output passes `ValidateRaw`: hallucinated IDs, zero-confidence verdicts, and malformed JSON degrade to `UNRESOLVED` via a single fallback path — never a silent `MATCHED`.
- 5-component evidence scoring (amount, reference similarity, date, description, uniqueness) with a separate decision-confidence metric; four metrics reported together: match rate, value reconciled, AI escalation rate, ground-truth accuracy.

HashVault — Cloud Storage Platform | *Go, Gin, PostgreSQL, Redis, RabbitMQ, AWS S3, Docker* — Mar 2026 – Present

- SHA-256 content-addressable deduplication: identical bytes map to one S3 key; reference counts are decremented atomically and the object is purged only when count reaches zero. Validated on **50 GB / 10K files**: **36% storage saved**, dedup-hit **<50 ms p95**, upload init **1K+ req/s**, API **<100 ms p95**.
- Two-token auth: HS256 JWT (15-min TTL) + 64-char opaque refresh tokens stored as SHA-256 hashes in Redis; prior token deleted on every rotation so stolen tokens cannot be replayed.
- Presigned S3 URLs keep file bytes off the Go server; async RabbitMQ pipeline (durable exchange, prefetch-1, ack/nack); Google OAuth 2.0 with CSRF state tokens; graceful 30-second SIGTERM shutdown.

Konnect — Real-Time Chat Platform | *Next.js, Express, MongoDB, Redis, RabbitMQ, Kafka, Docker* — Sept 2025

- 6-microservice architecture with Socket.IO gateway, Traefik reverse proxy, JWT RS256 dual-token auth, OTP email verification, and Google/GitHub OAuth 2.0.
- RabbitMQ + Kafka pub/sub backbone with Redis sliding-window aggregation for AI moderation; optimisation delivered **3.4× throughput** (320→1,090 msg/s), **71% P95 latency reduction** (340→98 ms), **49% CPU drop**, **62% fewer DB queries/msg**.
- Scales to **10K+ concurrent connections**, **1K+ msg/s**, **<100 ms** delivery, **99.9% success rate**, **<500 ms** moderation pipeline; **100K+ messages** processed. Published at IJSREM Vol. 10 (DOI: 10.55041/IJSREM61235).

Certifications

Developing Back-End Apps with Node.js and Express | *IBM / Coursera* — Jan 2026